

Codebook

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1 basic

1.1 build

```

1 code tem.cpp
2 ...
3 for i in {A..M}; do cp tem.cpp $i.cpp; done;
4 code *.cpp
5 code run
6
7 寫入以下編譯與執行指令至 run 檔案中：
8 g++ -std=c++20 -DABS -Wall -Wextra -Wshadow $1.cpp -o $1 && ./
9 $1
10 chmod +x run

```

1.2 default

```

1 #include <bits/stdc++.h>
2 #define int long long
3 #define pii pair<int, int>
4 #define F first
5 #define S second
6 #define pb push_back
7 #define all(x) x.begin(), x.end()
8 #define sz(x) (int)x.size()
9 #define FOR(i, a, b) for(int i = a; i <= b; ++i)
10 #define IO ios::sync_with_stdio(0), cin.tie(0)
11 using namespace std;
12 void debug() { cerr << endl; }
13 template <typename T, typename ...U>
14 void debug(T i, U ...j) { cerr << i << ' ', debug(j...); }
15 #define test(...) debug("[ " #__VA_ARGS__ " ]:", __VA_ARGS__)
16
17 signed main() {
18     IO;
19 }
20

```

1.3 div

```

1 ll floor(ll a, ll b) {
2     return a / b - (a < 0 && a % b);
3 }
4 ll ceil(ll a, ll b) {
5     return a / b + (a > 0 && a % b);
6 }
7 a / b < x -> floor(a, b) + 1 <= x
8 a / b <= x -> ceil(a, b) <= x
9 x < a / b -> x <= ceil(a, b) - 1
10 x <= a / b -> x <= floor(a, b)

```

1.4 fast

```

1 #pragma GCC optimize("Ofast,inline,unroll-loops")
2 #pragma GCC target("bmi,bmi2,lzcnt,popcnt,avx2")
3 #define windows_system
4 #ifdef windows_system
5     /* For Windows */
6 inline int getchar_unlocked() { return getchar(); }
7 inline void putchar_unlocked(char _c) { putchar(_c); }
8 #endif
9 inline int in() {
10     int re = 0;
11     char c = getchar_unlocked();
12     while (c == ' ' || c == '\n') c = getchar_unlocked();
13     while (c >= '0' && c <= '9') re = (re << 3) + (re << 1) + c - '0', c = getchar_unlocked();
14     return re;
15 }
16 inline void out(int x) {
17     char str[20];
18     int pos = 0;
19     do {
20         str[pos++] = x % 10 + '0';
21         x /= 10;
22     } while (x);
23     for (int i = pos - 1; i >= 0; i--) putchar_unlocked(str[i]);
24     putchar_unlocked('\n');
25 }

```

1.5 random

```

1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2     count());
3 inline int rnd(int l, int r) {
4     return uniform_int_distribution<int>(l, r)(seed);
5 }
6 inline double drnd(double l, double r) {
7     return uniform_real_distribution<double>(l, r)(seed);
8 }

```

2 data structure

2.1 2d seg

```

1 struct Node {
2     int x;
3     Node(int _x = 0) : x(_x) {}
4 };
5 Node combine(Node a, Node b) {
6     return Node(max(a.x, b.x));
7 }
8 struct Segment_Tree {
9     struct Segment_Tree2 {
10         vector<Node> seg2;
11         int n;
12         Segment_Tree2(int _n) : n(_n), seg2(4 * _n + 5) {}
13         void update2(int node, int l, int r, int id, int va) {
14             if (l == r) {
15                 seg2[node].x = va;
16                 return;
17             }
18             int mid = (l + r) >> 1;
19             if (id <= mid)
20                 update2((node << 1), l, mid, id, va);
21             else
22                 update2(((node << 1) | 1), mid + 1, r, id, va);
23             seg2[node] = combine(seg2[(node << 1)], seg2[(node << 1) | 1]);
24         }
25         Node query2(int node, int l, int r, int ql, int qr) {
26             if (ql <= l && r <= qr) return seg2[node];
27             int mid = (l + r) >> 1;
28             if (qr <= mid)
29                 return query2((node << 1), l, mid, ql, qr);
30             else if (mid + 1 <= ql)
31                 return query2(((node << 1) | 1), mid + 1, r, ql, qr);
32             else
33                 return combine(query2((node << 1), l, mid, ql, qr),
34                     query2(((node << 1) | 1), mid + 1, r, ql, qr));
35         }
36     };
37     vector<Segment_Tree2> seg;
38     int n;
39     Segment_Tree(int _n) : n(_n), seg(4 * _n + 5, Segment_Tree2(_n)) {}
40     void update(int node, int l, int r, int x, int y, int va) {
41         seg[node].update2(1, 1, n, y, va);
42         if (l == r) return;
43         int mid = (l + r) >> 1;
44         if (x <= mid)
45             update((node << 1), l, mid, x, y, va);
46         else
47             update(((node << 1) | 1), mid + 1, r, x, y, va);
48     }
49     Node query(int node, int l, int r, int xl, int xr, int yl,
50         int yr) {
51         if (xl <= l && r <= xr) return seg[node].query2(1, 1, n, yl, yr);
52         int mid = (l + r) >> 1;
53         if (xr <= mid)
54             return query((node << 1), l, mid, xl, xr, yl, yr);
55         else if (mid + 1 <= xl)
56             return query(((node << 1) | 1), mid + 1, r, xl, xr, yl, yr);
57         else
58             return combine(query((node << 1), l, mid, xl, xr, yl, yr),
59                 query(((node << 1) | 1), mid + 1, r, xl, xr, yl, yr));
60     }
61 }

```

2.2 2d seg ptr

```

1 struct Node2 {
2     int val;
3     Node2 *lc, *rc;
4     Node2 () {
5         val = 0;
6     }
7 };

```

```

5     lc = rc = nullptr;
6 }
7 };
8 struct Node1 {
9     Node2 *c; Node1 *lc, *rc;
10    Node1 () {
11        c = nullptr;
12        lc = rc = nullptr;
13    }
14 };
15 int N;
16 int Val2(Node2 *n2) { return (n2 ? n2->val : 011); }
17 void pull2(Node2 *n2) { n2->val = Val2(n2->lc) + Val2(n2->rc);
18 }
19 void update2(Node2 *n2, int l, int r, int id, int v) {
20     if (l == r) {
21         n2->val += v; return;
22     }
23     int mid = (l + r) >> 1;
24     if (id <= mid) {
25         if (!n2->lc) n2->lc = new Node2();
26         update2(n2->lc, l, mid, id, v);
27     } else {
28         if (!n2->rc) n2->rc = new Node2();
29         update2(n2->rc, mid + 1, r, id, v);
30     }
31     pull2(n2);
32 }
33 void update1(Node1 *n1, int l, int r, int qx, int qy, int v) {
34     if (!n1->c) n1->c = new Node2();
35     update2(n1->c, l, N, qy, v);
36     if (l == r) return;
37     int mid = (l + r) >> 1;
38     if (qx <= mid) {
39         if (!n1->lc) n1->lc = new Node1();
40         update1(n1->lc, l, mid, qx, qy, v);
41     } else {
42         if (!n1->rc) n1->rc = new Node1();
43         update1(n1->rc, mid + 1, r, qx, qy, v);
44     }
45 }
46 int query2(Node2 *n2, int l, int r, int ql, int qr) {
47     if (!n2) return 0;
48     if (ql <= l && r <= qr) return n2->val;
49     int mid = (l + r) >> 1;
50     if (qr <= mid)
51         return query2(n2->lc, l, mid, ql, qr);
52     else if (mid + 1 <= ql)
53         return query2(n2->rc, mid + 1, r, ql, qr);
54     else
55         return query2(n2->lc, l, mid, ql, qr) + query2(n2->rc, mid
56         + 1, r, ql, qr);
57 }
58 int query1(Node1 *n1, int l, int r, int qxl, int qxr, int qyl,
59 int qyr) {
60     if (!n1) return 0;
61     if (qxl <= l && r <= qxr) return query2(n1->c, l, N, qyl, qyr);
62     int mid = (l + r) >> 1;
63     if (qxr <= mid)
64         return query1(n1->lc, l, mid, qxl, qxr, qyl, qyr);
65     else if (mid + 1 <= qxl)
66         return query1(n1->rc, mid + 1, r, qxl, qxr, qyl, qyr);
67     else
68         return query1(n1->lc, l, mid, qxl, qxr, qyl, qyr) + query1(
69         n1->rc, mid + 1, r, qxl, qxr, qyl, qyr);
70 }

```

2.3 bit

```

1 struct BIT {
2     int N;
3     vector<int> info;
4     BIT(int n) : N(n), info(n + 1) {}
5     void update(int idx, int val) {
6         for (int i = idx; i <= N; i += i & -i)
7             info[i] += val;
8     }
9     int query(int idx) {
10        int res = 0;
11        for (int i = idx; i > 0; i -= i & -i)
12            res += info[i];
13        return res;
14    }
15    int kth(int k) { // kth small
16        int res = 0;
17        for (int i = 1 << __lg(N); i > 0; i >= 1) {
18            if (res + i <= N && info[res + i] < k) {
19                res += i;
20                k -= info[res];
21            }
22        }
23        return res + 1;
24    }

```

```
25 };
```

2.4 bitset

```

1 #pragma GCC target("popcnt")
2 // 15 = 1111
3 bitset<10> bit(15); // 宣告長度 = 10 的 bitset
4 cout << bit.count() << "\n"; // 回傳幾個 1
5 cout << bit.none() << "\n"; // 回傳是否全為 0
6 cout << bit.any() << "\n"; // 回傳是否有 1
7 cout << bit._Find_first() << "\n"; // 回傳第一個 bit = 1 的位
8 // 置
9 int x = 2;
10 cout << bit._Find_next(x) << "\n"; // 回傳 x 後 · 第一個 bit =
11 // 1 的位置

```

2.5 lichao

```

1 // MIN LeeChao Segment Tree
2 struct line {
3     int m, k;
4     line *lc, *rc;
5     line() { m = 0, k = INT_MAX, lc = nullptr, rc = nullptr; }
6     line(int _m, int _k) { m = _m, k = _k, lc = nullptr, rc =
7         nullptr; }
8     int calc(int x) {
9         return m * x + k;
10    }
11    void push(int l, int r) {
12        if (l == r) return;
13        if (!lc) lc = new line();
14        if (!rc) rc = new line();
15    }
16 };
17 struct Lee_Chao {
18     line* root;
19     Lee_Chao() : root(new line()) {}
20     void update(line* node, int l, int r, line add) {
21         if (l == r) {
22             if (add.calc(l) < node->calc(l)) {
23                 swap(add.m, node->m);
24                 swap(add.k, node->k);
25             }
26             return;
27         }
28         // 預設 seg[node] 的斜率 < add 的斜率
29         if (node->m > add.m) {
30             swap(add.m, node->m);
31             swap(add.k, node->k);
32         }
33         node->push(l, r);
34         int mid = (l + r) >> 1;
35         // 將小的 f(x) 留下
36         if (node->calc(mid) > add.calc(mid)) {
37             swap(add.m, node->m);
38             swap(add.k, node->k);
39             update(node->rc, mid + 1, r, add);
40         } else {
41             update(node->lc, l, mid, add);
42         }
43     }
44     int query(line* node, int l, int r, int x) {
45         if (node->k == INT_MAX) return INT_MAX;
46         if (l == r) return node->calc(x);
47         int mid = (l + r) >> 1;
48         if (x <= mid)
49             return min(node->calc(x), query(node->lc, l, mid, x));
50         else
51             return min(node->calc(x), query(node->rc, mid + 1, r, x));
52     }
53     void update(line l) { update(root, 1, N, l); }
54     int query(int x) { return query(root, 1, N, x); }
55 };

```

2.6 pbds

```

1 #include <ext/pb_ds/assoc_container.hpp>
2 #include <ext/pb_ds/tree_policy.hpp>
3 using namespace __gnu_pbds;
4 void solve() {
5     tree<int, null_type, greater<int>, rb_tree_tag,
6         tree_order_statistics_node_update> bst;
7     bst.insert(1);
8     int a = *bst.find_by_order(2); // 第 k + 1 大的數
9     int b = bst.order_of_key(2); // 有幾個數 > k
10    // -----

```

```

10 tree<int, null_type, less<int>, rb_tree_tag,
    tree_order_statistics_node_update> bst;
11 bst.insert(1);
12 int a = *bst.find_by_order(2); // 第 k + 1 小的數
13 int b = bst.order_of_key(2); // 有幾個數 < k
14 }

```

2.7 range set

```

1 struct RangeSet { // [L, r)
2     set<pii> st;
3     void cut(int x) {
4         auto it = st.lower_bound({x + 1, -1});
5         if (it == st.begin()) return;
6         auto [l, r] = *prev(it);
7         if (l >= x || x >= r) return;
8         st.erase(prev(it));
9         st.insert({l, x});
10        st.insert({x, r});
11    }
12    vector<pii> split(int l, int r) {
13        // remove and return ranges in [L, r)
14        cut(l), cut(r);
15        vector<pii> res;
16        while (true) {
17            auto it = st.lower_bound({l, -1});
18            if (it == st.end() || r <= it->first) break;
19            res.push_back(*it), st.erase(it);
20        }
21        return res;
22    }
23    void insert(int l, int r) {
24        // add a range [L, r), [L, r) not in st
25        auto it = st.lower_bound({l, r});
26        if (it != st.begin() && prev(it)->second == l)
27            l = prev(it)->first, st.erase(prev(it));
28        if (it != st.end() && r == it->first)
29            r = it->second, st.erase(it);
30        st.insert({l, r});
31    }
32    bool count(int x) {
33        auto it = st.lower_bound({x + 1, -1});
34        return it != st.begin() && prev(it)->first <= x && x < prev
            (it)->second;
35    }
36 };

```

2.8 sparse table

```

1 int a[N];
2 struct Sparse_table {
3     int n;
4     vector<vector<int>> st;
5     Sparse_table(int _n) : n(_n), st(_n + 1, vector<int>(__lg(_n)
6         + 1)) {
7         for (int i = 1; i <= n; i++) st[i][0] = a[i];
8         for (int lg = 1; lg <= __lg(n); lg++) {
9             int len = 1 << (lg - 1);
10            for (int i = 1; i + len <= n; i++)
11                st[i][lg] = max(st[i][lg - 1], st[i + len][lg - 1]);
12        }
13    }
14    int query(int l, int r) {
15        int lg = __lg(r - l + 1);
16        int len = 1 << lg;
17        return max(st[l][lg], st[r - len + 1][lg]);
18    }
19 };

```

2.9 stl sort

```

1 struct Data {
2     int x, y;
3 };
4 // priority_queue
5 struct cmp {
6     bool operator()(Data a, Data b) {
7         return a.x < b.x;
8     }
9 };
10 priority_queue<Data, vector<Data>, cmp> pq;
11 // set
12 struct cmp {
13     bool operator()(Data p, Data q) {
14         return p.x < q.x;
15     }
16 };
17 set<Data, cmp> st;

```

2.10 treap

```

1 struct node {
2     int data, sz;
3     node *l, *r;
4     node(int k) : data(k), sz(1), l(0), r(0) {}
5     void up() {
6         sz = 1;
7         if (l) sz += l->sz;
8         if (r) sz += r->sz;
9     }
10    void down() {}
11 };
12 int sz(node *a) { return a ? a->sz : 0; }
13 node *merge(node *a, node *b) {
14     if (!a || !b) return a ? a : b;
15     if (rand() % (sz(a) + sz(b)) < sz(a))
16         return a->down(), a->r = merge(a->r, b), a->up(), a;
17     return b->down(), b->l = merge(a, b->l), b->up(), b;
18 }
19 void split1(node *o, node *&a, node *&b, int k) {
20     if (!o) return a = b = 0, void();
21     o->down();
22     if (o->data <= k)
23         a = o, split1(o->r, a->r, b, k), a->up();
24     else
25         b = o, split1(o->l, a, b->l, k), b->up();
26 }
27 void split2(node *o, node *&a, node *&b, int k) {
28     if (sz(o) <= k) return a = o, b = 0, void();
29     o->down();
30     if (sz(o->l) + 1 <= k)
31         a = o, split2(o->r, a->r, b, k - sz(o->l) - 1);
32     else
33         b = o, split2(o->l, a, b->l, k);
34     o->up();
35 }
36 node *kth(node *o, int k) {
37     if (k <= sz(o->l)) return kth(o->l, k);
38     if (k == sz(o->l) + 1) return o;
39     return kth(o->r, k - sz(o->l) - 1);
40 }
41 int Rank(node *o, int key) {
42     if (!o) return 0;
43     if (o->data < key)
44         return sz(o->l) + 1 + Rank(o->r, key);
45     else
46         return Rank(o->l, key);
47 }
48 bool erase(node *&o, int k) {
49     if (!o) return 0;
50     if (o->data == k) {
51         node *t = o;
52         o->down(), o = merge(o->l, o->r);
53         delete t;
54         return 1;
55     }
56     node *t = k < o->data ? o->l : o->r;
57     return erase(t, k) ? o->up(), 1 : 0;
58 }
59 void insert(node *&o, int k) {
60     node *a, *b;
61     split(o, a, b, k),
62     o = merge(a, merge(new node(k), b));
63 }
64 void interval(node *&o, int l, int r) {
65     node *a, *b, *c;
66     split2(o, a, b, l - 1), split2(b, b, c, r);
67     // operate
68     o = merge(a, merge(b, c));
69 }

```

2.11 undo dsu

```

1 struct DSU {
2     vector<int> parent, size, tag;
3     vector<tuple<int, int, int, int, int>> history;
4     DSU(int n) {
5         parent.resize(n + 1);
6         size.resize(n + 1, 1);
7         tag.resize(n + 1, 0);
8         iota(all(parent), 0);
9     }
10    int find(int x) {
11        if (parent[x] == x) return x;
12        return find(parent[x]);
13    }
14    bool merge(int x, int y) {
15        x = find(x), y = find(y);
16        if (x == y) return false;
17        if (size[x] < size[y]) swap(x, y);

```

```

18     history.pb({x, y, parent[y], tag[x], size[x]});
19     parent[y] = x;
20     size[x] += size[y];
21     return true;
22 }
23 void add_tag(int head, int val) {
24     tag[head] += val;
25 }
26 void undo() {
27     auto [px, py, parent_py, base, sz_x] = history.back();
28     parent[py] = parent_py;
29     size[px] = sz_x;
30     int delta = tag[px] - base;
31     tag[py] += delta;
32     history.pop_back();
33 }
34 };

```

2.12 unordered map

```

1 struct custom_hash {
2     static uint64_t splitmix64(uint64_t x) {
3         x += 0x9e3779b97f4a7c15;
4         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
5         x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
6         return x ^ (x >> 31);
7     }
8     size_t operator()(uint64_t a) const {
9         static const uint64_t FIXED_RANDOM = chrono::steady_clock::
10             now().time_since_epoch().count();
11         return splitmix64(i + FIXED_RANDOM);
12     }
13 };
14 unordered_map<int, int, custom_hash> m1;

```

3 graph

3.1 2SAT

```

1 struct TWO_SAT {
2     int n, N;
3     vector<vector<int>>> G, rev_G;
4     deque<bool> used;
5     vector<int> order, comp;
6     deque<bool> assignment;
7     void init(int _n) {
8         n = _n;
9         N = _n * 2;
10        G.resize(N + 5);
11        rev_G.resize(N + 5);
12    }
13    void dfs1(int v) {
14        used[v] = true;
15        for (int u : G[v]) {
16            if (!used[u])
17                dfs1(u);
18        }
19        order.push_back(v);
20    }
21    void dfs2(int v, int cl) {
22        comp[v] = cl;
23        for (int u : rev_G[v]) {
24            if (comp[u] == -1)
25                dfs2(u, cl);
26        }
27    }
28    bool solve() {
29        order.clear();
30        used.assign(N, false);
31        for (int i = 0; i < N; ++i) {
32            if (!used[i])
33                dfs1(i);
34        }
35        comp.assign(N, -1);
36        for (int i = 0, j = 0; i < N; ++i) {
37            int v = order[N - i - 1];
38            if (comp[v] == -1)
39                dfs2(v, j++);
40        }
41        assignment.assign(n, false);
42        for (int i = 0; i < N; i += 2) {
43            if (comp[i] == comp[i + 1])
44                return false;
45            assignment[i / 2] = (comp[i] > comp[i + 1]);
46        }
47        return true;
48    }
49    void add_disjunction(int a, bool na, int b, bool nb) { // A
50        // or B

```

```

50        // na means whether a is negative or not
51        // nb means whether b is negative or not
52        a = 2 * a ^ na;
53        b = 2 * b ^ nb;
54        int neg_a = a ^ 1;
55        int neg_b = b ^ 1;
56        G[neg_a].push_back(b);
57        G[neg_b].push_back(a);
58        rev_G[b].push_back(neg_a);
59        rev_G[a].push_back(neg_b);
60        return;
61    }
62    void get_result(vector<int>& res) {
63        res.clear();
64        for (int i = 0; i < n; i++)
65            res.push_back(assignment[i]);
66    }
67 };

```

3.2 bridge

```

1 const int N = 200005;
2 vector<int> G[N];
3 int low[N], depth[N];
4 bool vis[N];
5 vector<pair<int, int>> bridge;
6 void dfs(int v, int p) {
7     vis[v] = true;
8     low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
9     for (int u : G[v]) {
10        if (u == p) continue;
11        if (!vis[u]) {
12            // (v, u) 是樹邊
13            dfs(u, v);
14            low[v] = min(low[v], low[u]);
15        } else {
16            // (v, u) 是回邊
17            low[v] = min(low[v], depth[u]);
18        }
19    }
20    // v 在不依靠父邊的情況下永遠沒辦法走到它的祖先
21    // (root, -1) 不算
22    if (low[v] == depth[v] && p != -1)
23        bridge.push_back({v, p});
24 }

```

3.3 bridge bcc

```

1 const int N = 200005;
2 vector<int> G[N];
3 int low[N], depth[N];
4 bool vis[N];
5 vector<vector<int>>> bcc;
6 stack<int> stk;
7 void dfs(int v, int p) {
8     stk.push(v);
9     vis[v] = true;
10    low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
11    for (int u : G[v]) {
12        if (u == p) continue;
13        if (!vis[u]) {
14            // (v, u) 是樹邊
15            dfs(u, v);
16            low[v] = min(low[v], low[u]);
17        } else {
18            // (v, u) 是回邊
19            low[v] = min(low[v], depth[u]);
20        }
21    }
22    // v 在不依靠父邊的情況下永遠沒辦法走到它的祖先
23    if (low[v] == depth[v]) {
24        bcc.emplace_back();
25        while (stk.top() != v) {
26            bcc.back().push_back(stk.top());
27            stk.pop();
28        }
29        bcc.back().push_back(stk.top());
30        stk.pop();
31    }
32 }

```

3.4 cut

```

1 const int N = 200005;
2 vector<int> G[N];
3 int low[N], depth[N];
4 bool vis[N], cut[N];
5 void dfs(int v, int p) {

```

```

6 vis[v] = true;
7 int child = 0;
8 low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
9 for (int u : G[v]) {
10     if (u == p) continue;
11     if (!vis[u]) {
12         // (v, u) 是樹邊
13         dfs(u, v);
14         child++;
15         low[v] = min(low[v], low[u]);
16         // u 無法在不經過父邊的情況走到 v 的祖先
17         if (low[u] >= depth[v] && p != -1)
18             cut[v] = true;
19     } else {
20         // (v, u) 是回邊
21         low[v] = min(low[v], depth[u]);
22     }
23 }
24 // 根節點有超過 2 個子節點
25 if (p == -1 && child >= 2)
26     cut[v] = true;
27 }

```

3.5 cut bcc

```

1 const int N = 200005;
2 vector<int> G[N];
3 int low[N], depth[N];
4 bool vis[N];
5 vector<vector<int>> bcc;
6 stack<int> stk;
7 void dfs(int v, int p) {
8     stk.push(v);
9     vis[v] = true;
10    low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
11    for (int u : G[v]) {
12        if (u == p) continue;
13        if (!vis[u]) {
14            // (v, u) 是樹邊
15            dfs(u, v);
16            low[v] = min(low[v], low[u]);
17            // u 無法在不經過父邊的情況走到 v 的祖先
18            if (low[u] >= depth[v]) {
19                bcc.emplace_back();
20                while (stk.top() != u) {
21                    bcc.back().push_back(stk.top());
22                    stk.pop();
23                }
24                bcc.back().push_back(stk.top());
25                stk.pop();
26                bcc.back().push_back(v);
27            }
28        } else {
29            // (v, u) 是回邊
30            low[v] = min(low[v], depth[u]);
31        }
32    }
33 }

```

3.6 kosaraju

```

1 const int N = 200005;
2 vector<int> G[N], rev_G[N], order;
3 bool vis[N]; int scc[N];
4 void rev_dfs(int v) {
5     vis[v] = true;
6     for (int u : rev_G[v])
7         if (!vis[u])
8             rev_dfs(u);
9     order.push_back(v);
10 }
11 void dfs(int v, int block) {
12     scc[v] = block;
13     for (int u : G[v])
14         if (scc[u] == -1)
15             dfs(u, block);
16 }
17 void kosaraju(int n) {
18     for (int i = 1; i <= n; i++)
19         if (!vis[i])
20             rev_dfs(i);
21     int scc_cnt = 0;
22     fill(scc, scc + N, -1);
23     for (int i = n - 1; i >= 0; i--)
24         if (scc[order[i]] == -1)
25             dfs(order[i], ++scc_cnt);
26 }

```

3.7 matching dfs

```

1 const int N = 2005;
2 vector<int> G[N];
3 struct Matching {
4     int n;
5     vector<bool> vis; vector<int> match;
6     Matching(int _n) : n(_n), vis(_n + 1), match(_n + 1, -1) {}
7     bool dfs(int v) {
8         vis[v] = true;
9         for (int u : G[v])
10             if (match[u] == -1 || (!vis[match[u]] && dfs(match[u])))
11                 {
12                     match[u] = v;
13                     return true;
14                 }
15         return false;
16     }
17     int solve() {
18         int cnt = 0;
19         for (int i = 1; i <= n; i++) {
20             vis.assign(n + 1, 0);
21             if (dfs(i)) cnt++;
22         }
23         return cnt;
24     }
25 };

```

3.8 matching general

```

1 // O(V^3)
2 // if graph.match[i] == n, node i is not matched
3 // else, i is matched with graph.match[i]
4 struct Matching { // 0-based
5     int fa[N], pre[N], match[N], s[N], v[N], n, tk;
6     vector<int> g[N];
7     queue<int> q;
8     int Find(int u) {
9         return u == fa[u] ? u : fa[u] = Find(fa[u]);
10    }
11    int lca(int x, int y) {
12        tk++;
13        x = Find(x), y = Find(y);
14        for (; ; swap(x, y)) {
15            if (x != n) {
16                if (v[x] == tk) return x;
17                v[x] = tk;
18                x = Find(pre[match[x]]);
19            }
20        }
21    }
22    void blossom(int x, int y, int l) {
23        while (Find(x) != l) {
24            pre[x] = y, y = match[x];
25            if (s[y] == 1) q.push(y), s[y] = 0;
26            if (fa[x] == x) fa[x] = l;
27            if (fa[y] == y) fa[y] = l;
28            x = pre[y];
29        }
30    }
31    bool bfs(int r) {
32        for (int i = 0; i <= n; ++i) fa[i] = i, s[i] = -1;
33        while (!q.empty()) q.pop();
34        q.push(r);
35        s[r] = 0;
36        while (!q.empty()) {
37            int x = q.front(); q.pop();
38            for (int u : g[x]) {
39                if (s[u] == -1) {
40                    pre[u] = x, s[u] = 1;
41                    if (match[u] == n) {
42                        for (int a = u, b = x, last; b != n; a = last, b = pre[a])
43                            last = match[b], match[b] = a, match[a] = b;
44                        return true;
45                    }
46                    q.push(match[u]);
47                    s[match[u]] = 0;
48                } else if (!s[u] && Find(u) != Find(x)) {
49                    int l = lca(u, x);
50                    blossom(x, u, l);
51                    blossom(u, x, l);
52                }
53            }
54        }
55        return false;
56    }
57    int solve() {
58        int res = 0;
59        for (int x = 0; x < n; ++x) {
60            if (match[x] == n) res += bfs(x);
61        }
62        return res;
63    }

```

```

63 }
64 void init(int _n) {
65     n = _n, tk = 0;
66     for (int i = 0; i <= n; ++i) match[i] = pre[i] = n;
67     for (int i = 0; i < n; ++i) g[i].clear(), v[i] = 0;
68 }
69 void add_edge(int u, int v) {
70     g[u].push_back(v), g[v].push_back(u);
71 }
72 };

```

3.9 matching general weight

```

1 // O(V^3)
2 // if graph.match[i] == 0, node i is not matched
3 // else, i is matched with graph.match[i]
4 struct WeightGraph { // 1-based
5     static const int inf = INT_MAX;
6     static const int maxn = 514;
7     struct edge {
8         int u, v, w;
9         edge(){}
10        edge(int u, int v, int w): u(u), v(v), w(w) {}
11    };
12    int n, n_x;
13    edge g[maxn * 2][maxn * 2];
14    int lab[maxn * 2];
15    int match[maxn * 2], slack[maxn * 2], st[maxn * 2], pa[maxn * 2];
16    int flo_from[maxn * 2][maxn + 1], S[maxn * 2], vis[maxn * 2];
17    vector<int> flo[maxn * 2];
18    queue<int> q;
19    int e_delta(const edge &e) { return lab[e.u] + lab[e.v] - g[e.u][e.v].w * 2; }
20    void update_slack(int u, int x) { if (!slack[x] || e_delta(g[u][x]) < e_delta(g[slack[x]][x])) slack[x] = u; }
21    void set_slack(int x) {
22        slack[x] = 0;
23        for (int u = 1; u <= n; ++u)
24            if (g[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
25                update_slack(u, x);
26    }
27    void q_push(int x) {
28        if (x <= n) q.push(x);
29        else for (size_t i = 0; i < flo[x].size(); ++i) q_push(flo[x][i]);
30    }
31    void set_st(int x, int b) {
32        st[x] = b;
33        if (x > n) for (size_t i = 0; i < flo[x].size(); ++i)
34            set_st(flo[x][i], b);
35    }
36    int get_pr(int b, int xr) {
37        int pr = find(flo[b].begin(), flo[b].end(), xr) - flo[b].begin();
38        if (pr % 2 == 1) {
39            reverse(flo[b].begin() + 1, flo[b].end());
40            return (int)flo[b].size() - pr;
41        }
42        return pr;
43    }
44    void set_match(int u, int v) {
45        match[u] = g[u][v].v;
46        if (u <= n) return;
47        edge e = g[u][v];
48        int xr = flo_from[u][e.u], pr = get_pr(u, xr);
49        for (int i = 0; i < pr; ++i) set_match(flo[u][i], flo[u][i ^ 1]);
50        set_match(xr, v);
51        rotate(flo[u].begin(), flo[u].begin() + pr, flo[u].end());
52    }
53    void augment(int u, int v) {
54        for (; ; ) {
55            int xnv = st[match[u]];
56            set_match(u, v);
57            if (!xnv) return;
58            set_match(xnv, st[pa[xnv]]);
59            u = st[pa[xnv]], v = xnv;
60        }
61    }
62    int get_lca(int u, int v) {
63        static int t = 0;
64        for (++t; u || v; swap(u, v)) {
65            if (u == 0) continue;
66            if (vis[u] == t) return u;
67            vis[u] = t;
68            u = st[match[u]];
69            if (u) u = st[pa[u]];
70        }
71        return 0;
72    }
73    void add_blossom(int u, int lca, int v) {
74        int b = n + 1;
75        while (b <= n_x && st[b]) ++b;

```

```

75     if (b > n_x) ++n_x;
76     lab[b] = 0, S[b] = 0;
77     match[b] = match[lca];
78     flo[b].clear();
79     flo[b].push_back(lca);
80     for (int x = u, y; x != lca; x = st[pa[y]])
81         flo[b].push_back(x), flo[b].push_back(y = st[match[x]]),
82         q_push(y);
83     reverse(flo[b].begin() + 1, flo[b].end());
84     for (int x = v, y; x != lca; x = st[pa[y]])
85         flo[b].push_back(x), flo[b].push_back(y = st[match[x]]),
86         q_push(y);
87     set_st(b, b);
88     for (int x = 1; x <= n_x; ++x) g[b][x].w = g[x][b].w = 0;
89     for (int x = 1; x <= n; ++x) flo_from[b][x] = 0;
90     for (size_t i = 0; i < flo[b].size(); ++i) {
91         int xs = flo[b][i];
92         for (int x = 1; x <= n_x; ++x)
93             if (g[b][x].w == 0 || e_delta(g[xs][x]) < e_delta(g[b][x]))
94                 g[b][x] = g[xs][x], g[x][b] = g[x][xs];
95         for (int x = 1; x <= n; ++x)
96             if (flo_from[xs][x]) flo_from[b][x] = xs;
97     }
98     set_slack(b);
99 }
100 void expand_blossom(int b) {
101     for (size_t i = 0; i < flo[b].size(); ++i)
102         set_st(flo[b][i], flo[b][i]);
103     int xr = flo_from[b][g[b][pa[b]].u], pr = get_pr(b, xr);
104     for (int i = 0; i < pr; ++i) {
105         int xs = flo[b][i], xns = flo[b][i + 1];
106         pa[xs] = g[xns][xs].u;
107         S[xs] = 1, S[xns] = 0;
108         slack[xs] = 0, set_slack(xns);
109         q_push(xns);
110     }
111     S[xr] = 1, pa[xr] = pa[b];
112     for (size_t i = pr + 1; i < flo[b].size(); ++i) {
113         int xs = flo[b][i];
114         S[xs] = -1, set_slack(xs);
115     }
116     st[b] = 0;
117 }
118 bool on_found_edge(const edge &e) {
119     int u = st[e.u], v = st[e.v];
120     if (S[v] == -1) {
121         pa[v] = e.u, S[v] = 1;
122         int nu = st[match[v]];
123         slack[v] = slack[nu] = 0;
124         S[nu] = 0, q_push(nu);
125     } else if (S[v] == 0) {
126         int lca = get_lca(u, v);
127         if (!lca)
128             return augment(u, v), augment(v, u), true;
129         else
130             add_blossom(u, lca, v);
131     }
132     return false;
133 }
134 bool matching() {
135     memset(S + 1, -1, sizeof(int) * n_x);
136     memset(slack + 1, 0, sizeof(int) * n_x);
137     q = queue<int>();
138     for (int x = 1; x <= n_x; ++x)
139         if (st[x] == x && !match[x])
140             pa[x] = 0, S[x] = 0, q_push(x);
141     if (q.empty()) return false;
142     for (; ; ) {
143         while (q.size()) {
144             int u = q.front(); q.pop();
145             if (S[st[u]] == 1) continue;
146             for (int v = 1; v <= n; ++v)
147                 if (g[u][v].w > 0 && st[u] != st[v]) {
148                     if (e_delta(g[u][v]) == 0) {
149                         if (on_found_edge(g[u][v]))
150                             return true;
151                     } else
152                         update_slack(u, st[v]);
153                 }
154             }
155         int d = inf;
156         for (int b = n + 1; b <= n_x; ++b)
157             if (st[b] == b && S[b] == 1) d = min(d, lab[b] / 2);
158         for (int x = 1; x <= n_x; ++x)
159             if (st[x] == x && slack[x]) {
160                 if (S[x] == -1) d = min(d, e_delta(g[slack[x]][x]));
161                 else if (S[x] == 0) d = min(d, e_delta(g[slack[x]][x]) / 2);
162             }
163         for (int u = 1; u <= n; ++u) {
164             if (S[st[u]] == 0) {
165                 if (lab[u] <= d) return 0;
166                 lab[u] -= d;
167             } else if (S[st[u]] == 1) lab[u] += d;
168         }
169         for (int b = n + 1; b <= n_x; ++b)

```

```

168     if (st[b] == b) {
169         if (S[st[b]] == 0) lab[b] += d * 2;
170         else if (S[st[b]] == 1) lab[b] -= d * 2;
171     }
172     q = queue<int>();
173     for (int x = 1; x <= n_x; ++x)
174         if (st[x] == x && slack[x] && st[slack[x]] != x &&
175             e_delta(g[slack[x]][x]) == 0)
176             if (on_found_edge(g[slack[x]][x]))
177                 return true;
178     for (int b = n + 1; b <= n_x; ++b)
179         if (st[b] == b && S[b] == 1 && lab[b] == 0)
180             expand_blossom(b);
181 }
182 return false;
183 }
184 pair<long long, int> solve() {
185     memset(match + 1, 0, sizeof(int) * n);
186     n_x = n;
187     int n_matches = 0;
188     long long tot_weight = 0;
189     for (int u = 0; u <= n; ++u) st[u] = u, flo[u].clear();
190     int w_max = 0;
191     for (int u = 1; u <= n; ++u)
192         for (int v = 1; v <= n; ++v) {
193             flo_from[u][v] = (u == v ? u : 0);
194             w_max = max(w_max, g[u][v].w);
195         }
196     for (int u = 1; u <= n; ++u) lab[u] = w_max;
197     while (matching()) ++n_matches;
198     for (int u = 1; u <= n; ++u)
199         if (match[u] && match[u] < u)
200             tot_weight += g[u][match[u]].w;
201     return make_pair(tot_weight, n_matches);
202 }
203 void add_edge(int ui, int vi, int wi) {
204     g[ui][vi].w = g[vi][ui].w = wi;
205 }
206 void init(int _n) {
207     n = _n;
208     for (int u = 1; u <= n; ++u)
209         for (int v = 1; v <= n; ++v)
210             g[u][v] = edge(u, v, 0);
211 }
};

```

3.10 prim

```

1  const int N = 2005;
2  int dis[N][N];
3  int distfromtree[N];
4  for (int i = 1; i <= n; i++)
5      distfromtree[i] = dis[1][i];
6  int res = 0;
7  for (int i = 2; i <= n; i++) {
8      int mindist = INT_MAX, next = -1;
9      for (int j = 1; j <= n; j++) {
10         if (distfromtree[j] != 0 && distfromtree[j] < mindist) {
11             mindist = distfromtree[j];
12             next = j;
13         }
14     }
15     res += mindist;
16     for (int j = 1; j <= n; j++)
17         distfromtree[j] = min(distfromtree[j], dis[next][j]);
18 }

```

3.11 spfa

```

1  vector<int> G[N];
2  int spfa(int n, int s, int t) {
3      vector<int> dis(n, INT_MAX), cnt(n);
4      vector<bool> inq(n);
5      queue<int> q;
6      q.push(s);
7      dis[s] = 0;
8      inq[s] = true;
9      while (q.size()) {
10         int u = q.front();
11         q.pop();
12         cnt[u]++;
13         if (cnt[u] == n) // negative cycle
14             return INT_MAX;
15         inq[u] = false;
16         for (auto [v, w] : G[u]) {
17             if (dis[u] + w < dis[v]) {
18                 dis[v] = dis[u] + w;
19                 if (!inq[v]) {
20                     inq[v] = true;
21                     q.push(v);
22                 }

```

```

23         }
24     }
25     }
26     return dis[t];
27 }

```

4 math

4.1 big number

```

1  void init(string &a, string &b) {
2      while (a.size() < b.size()) a = '0' + a;
3      while (a.size() > b.size()) b = '0' + b;
4  }
5  int Compare(string &a, string &b) {
6      if (a < b) {
7          swap(a, b); return -1;
8      }
9      return 1;
10 }
11 bool del(string &a) {
12     if (a[0] == '0') {
13         a.erase(0, 1); return true;
14     }
15     return false;
16 }
17 void del_all_zero(string &a) {
18     while (del(a)) ;
19 }
20 string add(string a, string b) {
21     init(a, b);
22     a = '0' + a; b = '0' + b;
23     for (int i = a.size() - 1; i >= 0; i--) {
24         int p = a[i] - '0', q = b[i] - '0';
25         if (p + q >= 10) {
26             a[i - 1] = '0' + (a[i - 1] - '0' + 1);
27             a[i] = '0' + (p + q - 10);
28         } else {
29             a[i] = '0' + (p + q);
30         }
31     }
32     del(a);
33     return a;
34 }
35 string sub(string a, string b) {
36     init(a, b);
37     if (a == b) return "0";
38     int flag = Compare(a, b);
39     for (int i = a.size() - 1; i >= 0; i--) {
40         int p = a[i] - '0', q = b[i] - '0';
41         if (p < q) {
42             a[i - 1] = '0' + (a[i - 1] - '0' - 1);
43             a[i] = '0' + (p - q + 10);
44         } else {
45             a[i] = '0' + (p - q);
46         }
47     }
48     del_all_zero(a);
49     if (flag == -1) a = "-" + a;
50     return a;
51 }
52 string mul(string a, string b) {
53     string res = "0";
54     init(a, b);
55     Compare(a, b);
56     del_all_zero(b);
57     for (int i = b.size() - 1; i >= 0; i--) {
58         int x = b[i] - '0';
59         if (i != b.size() - 1) a = a + '0';
60         for (int i = 0; i < x; i++)
61             res = add(a, res);
62     }
63     return res;
64 }
65 string div(string a, string b) {
66     init(a, b);
67     if (a < b) return "0";
68     del_all_zero(b);
69     string res = "0", restmp = "1", tmp = b;
70     for (int i = 0; i < (a.size() - b.size()); i++) {
71         restmp += '0';
72         tmp += '0';
73     }
74     init(a, b);
75     while (a >= b) {
76         init(a, tmp);
77         if (a >= tmp) {
78             a = sub(a, tmp);
79             res = add(res, restmp);
80         } else {
81             restmp.erase(restmp.size() - 1);
82             tmp.erase(tmp.size() - 1);

```

```

83     }
84     init(a, b);
85 }
86 return res;
87 }

```

4.2 burnside lemma

$$\sum_{k=1}^n \frac{c(k)}{n}$$

- n : 有多少種置換方式 (例如: 旋轉方式)
- $c(k)$: 所有可能中 · 經過 k 次旋轉後 · 仍不會和別人相同方式的數量

4.3 catalan

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

4.4 count prime

```

1 // Count the number of primes not greater than N (N <= 10^11)
2 int primeCount(const int n) {
3     if (n <= 1) { return 0; }
4     if (n == 2) { return 1; }
5     const int v = sqrtl(n);
6     int s = (v + 1) / 2;
7     vector<int> smalls(s), roughs(s), skip(v + 1), larges(s);
8     iota(smalls.begin(), smalls.end(), 0);
9     for (int i = 0; i < s; i++) {
10         roughs[i] = 2 * i + 1;
11         larges[i] = (n / roughs[i] - 1) / 2;
12     }
13     const auto half = [](int n) -> int { return (n - 1) >> 1; };
14     int pc = 0;
15     for (int p = 3; p <= v; p += 2) {
16         if (skip[p]) { continue; }
17         int q = p * p;
18         if (1LL * q * q > n) { break; }
19         skip[p] = true;
20         for (int i = q; i <= v; i += 2 * p) skip[i] = true;
21         int ns = 0;
22         for (int k = 0; k < s; k++) {
23             int i = roughs[k];
24             if (skip[i]) { continue; }
25             int d = 1LL * i * p;
26             larges[ns] = larges[k] - (d <= v ? larges[smalls[d / 2] - pc] : smalls[half(n / d)]) + pc;
27             roughs[ns++] = i;
28         }
29         s = ns;
30         for (int i = half(v), j = v / p - 1 | 1; j >= p; j -= 2) {
31             int c = smalls[j / 2] - pc;
32             for (int e = j * p / 2; i >= e; i--) { smalls[i] -= c; }
33         }
34         pc++;
35     }
36     larges[0] += 1LL * (s + 2 * (pc - 1)) * (s - 1) / 2;
37     for (int k = 1; k < s; k++) { larges[0] -= larges[k]; }
38     for (int l = 1; l < s; l++) {
39         int q = roughs[l];
40         int M = n / q;
41         int e = smalls[half(M / q)] - pc;
42         if (e <= 1) { break; }
43         int t = 0;
44         for (int k = 1 + 1; k <= e; k++) { t += smalls[half(M / roughs[k])]; }
45         larges[0] += t - 1LL * (e - 1) * (pc + 1 - 1);
46     }
47     return larges[0] + 1;
48 }

```

4.5 CRT coprime

```

1 vector<int> a, m;
2 int extgcd(int a, int b, int& x, int& y) {
3     if (b == 0) {
4         x = 1, y = 0;
5         return a;
6     }
7     int ret = extgcd(b, a % b, y, x);
8     y -= a / b * x;
9     return ret;
10 }
11 // n = 有幾個式子 · 求解 x ≡ a_i (mod m_i)
12 int CRT(int n, vector<int>& a, vector<int>& m) {

```

```

13     int p = 1, ans = 0;
14     vector<int> M(n), inv_M(n);
15     for (int i = 0; i < n; i++)
16         p *= m[i];
17     for (int i = 0; i < n; i++) {
18         M[i] = p / m[i];
19         extgcd(M[i], m[i], inv_M[i], tmp);
20         ans += a[i] * inv_M[i] * M[i];
21         ans %= p;
22     }
23     return (ans % p + p) % p;
24 }

```

4.6 CRT coprime

n = 有幾個式子 · 求解 $x \equiv a_i \pmod{m_i}$

4.7 CRT not coprime

```

1 int extgcd(int a, int b, int& x, int& y) {
2     if (b == 0) {
3         x = 1, y = 0;
4         return a;
5     }
6     int ret = extgcd(b, a % b, y, x);
7     y -= a / b * x;
8     return ret;
9 }
10 // 對於方程組的式子兩兩求解 · return {是否有解, {a, m}}
11 pair<bool, pair<int, int>> CRT(int a1, int m1, int a2, int m2)
12 {
13     int g = __gcd(m1, m2);
14     if ((a2 - a1) % g != 0)
15         return {0, {-1, -1}};
16     int x, y;
17     extgcd(m1, m2, x, y);
18     x = (a2 - a1) * x / g; // 兩者不能相反
19     a1 = x * m1 + a1;
20     m1 = m1 * m2 / g;
21     a1 = (a1 % m1 + m1) % m1;
22     return {1, {a1, m1}};
23 }

```

4.8 det

```

1 ll Det(vector<vector<ll>> a) {
2     int n = a.size();
3     ll det = 1;
4     for (int i = 0; i < n; ++i) {
5         if (!a[i][i]) {
6             det = -det;
7             if (det < 0) det += mod;
8             for (int j = i + 1; j < n; ++j) if (a[j][i]) {
9                 swap(a[j], a[i]);
10                break;
11            }
12            if (!a[i][i]) return 0;
13        }
14        det = det * a[i][i] % mod;
15        ll mul = mpow(a[i][i], mod - 2);
16        for (int j = 0; j < n; ++j)
17            a[i][j] = a[i][j] * mul % mod;
18        for (int j = 0; j < n; ++j) if (i ^ j) {
19            ll mul = a[j][i];
20            for (int k = 0; k < n; ++k) {
21                a[j][k] -= a[i][k] * mul % mod;
22                if (a[j][k] < 0) a[j][k] += mod;
23            }
24        }
25    }
26    return det;
27 }

```

4.9 extgcd

```

1 // a * p.first + b * p.second = gcd(a, b)
2 pair<ll, ll> extgcd(ll a, ll b) {
3     pair<ll, ll> res, tmp;
4     ll f = 1, g = 1;
5     if (a < 0) a *= -1, f *= -1;
6     if (b < 0) b *= -1, g *= -1;
7     if (b == 0) return {f, 0};
8     tmp = extgcd(b, a % b);
9     res.first = tmp.second * f;
10    res.second = (tmp.first - tmp.second * (a / b)) * g;
11    return res;
12 }

```

4.10 factorize

```

1 using i64 = long long;
2 using i128 = __int128_t;
3 struct Factorize { // 質因數分解 N <=
4     i64 fmul(i64 a, i64 b, i64 p) {
5         return (i128)a * b % p;
6     }
7     i64 fpow(i64 a, i64 b, i64 p) {
8         i64 res = 1;
9         for (; b; b >>= 1, a = fmul(a, a, p))
10             if (b & 1) res = fmul(res, a, p);
11         return res;
12     }
13     bool check(i64 a, i64 u, i64 n, int t) {
14         a = fpow(a, u, n);
15         if (a == 0 || a == 1 || a == n - 1)
16             return true;
17         for (int i = 0; i < t; i++) {
18             a = fmul(a, a, n);
19             if (a == 1) return false;
20             if (a == n - 1) return true;
21         }
22         return false;
23     };
24     bool isPrime(i64 n) {
25         constexpr array<i64, 7> magic{2, 235, 9375, 28178, 450775,
26             9780504, 1795265022};
27         // for int: {2, 7, 61}
28         if (n < 2) return false;
29         if (n % 2 == 0) return n == 2;
30         i64 u = n - 1;
31         int t = 0;
32         while (u % 2 == 0) u >>= 1, t++;
33         for (auto v : magic) if (!check(v, u, n, t)) return false;
34         return true;
35     }
36     i64 PollardRho(i64 n) { // return non-trivial factor of n
37         if (n % 2 == 0) return 2;
38         i64 x = 2, y = 2, d = 1, p = 1;
39         auto f = [](i64 x, i64 n, i64 p) -> i64 {
40             return ((i128)x * x % n + p) % n;
41         };
42         while (true) {
43             x = f(x, n, p);
44             y = f(f(y, n, p), n, p);
45             d = gcd(abs(x - y), n);
46             if (d != n && d != 1) return d;
47             if (d == n) ++p;
48         }
49     }
50     i64 primeFactor(i64 n) {
51         return isPrime(n) ? n : primeFactor(PollardRho(n));
52     }
53 } ftr;
54 void solve() {
55     i64 n;
56     cin >> n;
57     vector<i64> ans;
58     while (n > 1) {
59         i64 p = ftr.primeFactor(n);
60         do {
61             ans.push_back(p);
62             n /= p;
63         } while (n % p == 0);
64     }
65     sort(ans.begin(), ans.end());
66     cout << ans.size() << ' ';
67     for (int i : ans) cout << i << ' ';
68     cout << '\n';
69 }

```

4.11 fraction

```

1 struct frac {
2     int a, b;
3     frac(int _a = 0, int _b = 1) : a(_a), b(_b) {
4         int g = __gcd(a, b);
5         a /= g; b /= g;
6         if (b < 0) { a *= -1; b *= -1; }
7     }
8 };
9 frac operator+(frac x, frac y) { return frac(x.a * y.b + y.a *
10     x.b, x.b * y.b); }
11 frac operator-(frac x, frac y) { return frac(x.a * y.b - y.a *
12     x.b, x.b * y.b); }
13 frac operator*(frac x, frac y) { return frac(x.a * y.a, x.b * y
14     .b); }
15 frac operator/(frac x, frac y) { return frac(x.a * y.b, x.b * y
16     .a); }
17 bool operator>(frac x, frac y) { return x.a * y.b > y.a * x.b;
18 }

```

```

14 bool operator<(frac x, frac y) { return x.a * y.b < y.a * x.b;
15 }
16 bool operator>=(frac x, frac y) { return x.a * y.b >= y.a * x.b
17 ; }
18 bool operator<=(frac x, frac y) { return x.a * y.b <= y.a * x.b
19 ; }
20 bool operator==(frac x, frac y) { return x.a * y.b == y.a * x.b
21 ; }
22 frac abs(frac x) { return frac(abs(x.a), abs(x.b)); }
23 ostream& operator<<(ostream& os, const frac& x) { os << x.a <<
24     "/" << x.b; return os; }

```

4.12 gaussian elimination

```

1 #define int long long
2 const int N = 25;
3 frac co[N][N];
4 // SIZE = n * (n + 1)
5 // a_1 x_1 + a_2 x_2 + ... + a_n x_n = a_{n+1}
6 void gaussian(int n) {
7     for (int i = 1; i <= n; i++) {
8         if (co[i][i].a == 0 && co[i][i].b == 1) { // swap
9             int swapid = -1;
10             for (int id = i + 1; id <= n; id++) {
11                 if (! (co[id][i].a == 0 && co[id][i].b == 1)) {
12                     swapid = id; break;
13                 }
14             }
15             if (swapid == -1)
16                 continue;
17             else {
18                 for (int j = 1; j <= n + 1; j++)
19                     swap(co[i][j], co[swapid][j]);
20             }
21         }
22         for (int id = i + 1; id <= n; id++) {
23             if (! (co[id][i].a == 0 && co[id][i].b == 1)) {
24                 frac mul = co[id][i] / co[i][i];
25                 for (int j = 1; j <= n + 1; j++)
26                     co[id][j] = co[id][j] - co[i][j] * mul;
27             }
28         }
29     }
30     for (int i = n; i >= 1; i--) {
31         if (! (co[i][i].a == 0 && co[i][i].b == 1)) {
32             co[i][n + 1] = co[i][n + 1] / co[i][i];
33             co[i][i] = {1, 1};
34         }
35         else
36             continue;
37         for (int id = 1; id < i; id++) {
38             if (! (co[id][i].a == 0 && co[id][i].b == 1)) {
39                 frac mul = co[id][i] / co[i][i];
40                 for (int j = 1; j <= n + 1; j++)
41                     co[id][j] = co[id][j] - co[i][j] * mul;
42             }
43         }
44     }
45     bool check_inf = false, check_no = false;
46     for (int i = 1; i <= n; i++) {
47         bool allzro = true;
48         for (int j = 1; j <= n; j++)
49             if (! (co[i][j].a == 0 && co[i][j].b == 1))
50                 allzro = false;
51         if (allzro) {
52             if (! (co[i][n + 1].a == 0 && co[i][n + 1].b == 1))
53                 check_no = true;
54             else
55                 check_inf = true;
56         }
57     }
58     if (check_no) {
59         cout << "no solutions\n";
60     }
61     else if (check_inf) {
62         cout << "infinite many solutions\n";
63     }
64     else {
65         for (int i = 1; i <= n; i++)
66             cout << "x" << i << " = " << co[i][n + 1] << '\n';
67     }
68 }

```

4.13 josephus

```

1 // 有 n 個人，第偶數個報數的人被刪掉，問第 k 個被踢掉的是誰
2 int solve(int n, int k) {
3     if (n == 1) return 1;
4     if (k <= (n + 1) / 2) {
5         if (2 * k > n)
6             return 2 * k % n;
7         else
8             return 2 * k;
9     }

```

```

9 | } else {
10 |     int res = solve(n / 2, k - (n + 1) / 2);
11 |     if (n & 1)
12 |         return 2 * res + 1;
13 |     else
14 |         return 2 * res - 1;
15 | }
16 | }

```

4.14 lucas

```

1 | // 對於很大的  $C^n_m$  對質數  $p$  取模，只要  $p$  不大就可以用
2 | int Lucas(int n, int m, int p) {
3 |     if (m == 0) return 1;
4 |     return (C(n % p, m % p, p) * Lucas(n / p, m / p, p) % p);
5 | }

```

4.15 math block

```

1 | // the sum of floor(n / i) for i = 1 ~ n
2 | int f(int n) {
3 |     int ans = 0;
4 |     int l = 1, r = 0;
5 |     while (l <= n) {
6 |         r = n / (n / l);
7 |         ans += (r - l + 1) * (n / l);
8 |         l = r + 1;
9 |     }
10 |    return ans;
11 | }
12 | // the sum of ceil(n / i) for i = 1 ~ n
13 | int f(int n) {
14 |     int ans = 0;
15 |     int l = 1, r = 0;
16 |     while (l < n) {
17 |         r = (n - 1) / ((n - 1) / l);
18 |         ans += (r - l + 1) * ((n + l - 1) / l);
19 |         l = r + 1;
20 |     }
21 |     if (l == n) ans += (n + l - 1) / l;
22 |     return ans;
23 | }

```

4.16 miller rabin

```

1 | struct Miller_Rabin {
2 |     static constexpr uint32_t ws32[3] = {2, 7, 61};
3 |     static constexpr uint64_t ws64[7] = {2, 325, 9375, 28178,
4 |         450775, 9780504, 1795265022};
5 |     template <class uint>
6 |     static constexpr uint fast_pow(uint b, uint p, uint mod) {
7 |         using Uint = conditional_t<is_same_v<uint, uint32_t>,
8 |             uint64_t, __uint128_t>;
9 |         uint ret{1};
10 |        for (; p; p >>= 1) {
11 |            if (p & 1) ret = Uint{ret} * b % mod;
12 |            b = Uint{b} * b % mod;
13 |        }
14 |        return ret;
15 |    }
16 |    template <class uint, class = enable_if_t<is_unsigned_v<uint>
17 |        >>>
18 |    static constexpr bool is_prime(uint n) {
19 |        const auto& witness = []() -> const auto& {
20 |            if constexpr (is_same_v<uint, uint32_t>) return ws32;
21 |            else return ws64;
22 |        }();
23 |        using Uint = conditional_t<is_same_v<uint, uint32_t>,
24 |            uint64_t, __uint128_t>;
25 |        if (n < 3 || !(n & 1)) return n == 2;
26 |        auto u = n - 1, t = (uint) __builtin_ctzll(u);
27 |        u >>= t;
28 |        for (auto x : witness) {
29 |            auto a = x % n;
30 |            if (a == 0 || a == 1 || a == n - 1) continue;
31 |            auto v = fast_pow(a, u, n);
32 |            uint i = 0;
33 |            for (; v != 1 && v != n - 1 && i < t; i++) v = Uint{v} *
34 |                v % n;
35 |            if (v != n - 1 && i > 0) return false;
36 |        }
37 |        return true;
38 |    }
39 |    template <class I, class = enable_if_t<is_integral_v<I>>>
40 |    constexpr bool operator()(I x) const {
41 |        if constexpr (is_signed_v<I>) if (x < 0) return false;
42 |        using U = make_unsigned_t<I>;

```

```

38 |     return is_prime((U)x);
39 | }
40 | } is_prime;

```

4.17 phi

```

1 | // phi(n) = 小於 n 並與 n 互質的正整數數量。
2 | // O(sqrt(n)) · 回傳 phi(n)
3 | int phi(int n) {
4 |     int ret = n;
5 |     for (int i = 2; i * i <= n; i++) {
6 |         if (n % i == 0) {
7 |             while (n % i == 0) n /= i;
8 |             ret = ret * (i - 1) / i;
9 |         }
10 |    }
11 |    if (n > 1) ret = ret * (n - 1) / n;
12 |    return ret;
13 | }
14 | // O(n log n) · 回傳 1 ~ n 的 phi 值
15 | vector<int> phi_1_to_n(int n) {
16 |     vector<int> phi(n + 1);
17 |     phi[0] = 0; phi[1] = 1;
18 |     for (int i = 2; i <= n; i++) phi[i] = i - 1;
19 |     for (int i = 2; i <= n; i++)
20 |         for (int j = 2 * i; j <= n; j += i) // 枚舉所有倍數
21 |             phi[j] -= phi[i];
22 |     return phi;
23 | }

```

4.18 phi lemma

若 a, m 互質，則：

$$a^n \bmod m = a^{n \bmod \varphi(m)} \bmod m$$

若 a, m 可能是任何數，則：

$$a^{\varphi(m) + [n \bmod \varphi(m)]} \bmod m$$

4.19 primes er

```

1 | // Time Complexity : O(nloglogn)
2 | const int N = 200005;
3 | bool is_prime[N];
4 | vector<int> primes;
5 | void build() {
6 |     fill(is_prime, is_prime + N, true);
7 |     is_prime[0] = is_prime[1] = false;
8 |     for (int i = 2; i <= 200000; i++) {
9 |         if (is_prime[i]) {
10 |            primes.push_back(i);
11 |            if (1ll * i * i <= 200000)
12 |                for (int j = i * i; j <= 200000; j += i)
13 |                    is_prime[j] = false;
14 |        }
15 |    }
16 | }

```

4.20 primes linear

```

1 | // Time Complexity : O(n)
2 | // if Lpf(i) = i, then it means that i is a prime.
3 | // else Lpf(i) is the smallest prime factor.
4 | // The 199999-th of the prime is 2750131
5 | bool is_prime[2750135];
6 | int lpf[2750135];
7 | vector<int> primes;
8 | void init() {
9 |     fill(is_prime, is_prime + 2750135, true);
10 |    for (int i = 2; i <= 2750131; i++) {
11 |        if (is_prime[i]) {
12 |            primes.push_back(i);
13 |            lpf[i] = i;
14 |        }
15 |        for (int p : primes) {
16 |            if (p * i > 2750131) break;
17 |            is_prime[p * i] = false;
18 |            lpf[p * i] = p;
19 |            if (i % p == 0) break;
20 |        }
21 |    }
22 | }

```

4.21 two equations

$$\begin{cases} ax + by = c \\ dx + ey = f \end{cases} = \begin{cases} x = \frac{ed-bf}{ad-bc} \\ y = \frac{af-ec}{ad-bc} \end{cases}$$

若 $x = \frac{0}{0}$ 且 $y = \frac{0}{0}$ ，則代表無限多組解。若 $x = \frac{*}{0}$ 且 $y = \frac{*}{0}$ ，則代表無解。

5 other

5.1 aliens dp

```

1 const int N = 2000005;
2 int n, k;
3 int a[N];
4 pii dp0[N], dp1[N];
5 pii calc(int c) {
6     dp0[1] = {0, 0}; dp1[1] = {c - a[1], 1};
7     for (int i = 2; i <= n; i++) {
8         dp0[i] = max(dp0[i - 1], dp1[i - 1] + make_pair(+a[i], 0));
9         dp1[i] = max(dp1[i - 1], dp0[i - 1] + make_pair(-a[i] + c,
10             1));
11     }
12     return dp0[n];
13 }
14 void solve() {
15     pii p = calc(0);
16     if (p.S <= k) {
17         cout << p.F << "\n";
18         return;
19     }
20     int l = -1e5, r = 1e6; pii res;
21     while (l + 1 < r) {
22         int mid = (l + r) >> 1;
23         res = calc(mid);
24         if (res.S == k) {
25             cout << res.F - mid * k << "\n";
26             return;
27         } else if (res.S > k) {
28             r = mid;
29         } else {
30             l = mid;
31         }
32     }
33     pii p1 = calc(l), p2 = calc(r);
34     int dp_1 = p1.F - l * p1.S;
35     int dp_2 = p2.F - r * p2.S;
36     cout << (dp_1 + (dp_2 - dp_1) * (k - p1.S) / (p2.S - p1.S))
37         << "\n";
38 }

```

5.2 dc dp

```

1 const int N = 3005;
2 int n, k, a[N], pre[N], dp[N][N];
3 int cost(int l, int r) {
4     void dc(int j, int l, int r, int best_l, int best_r) {
5         if (l > r) return;
6         int mid = (l + r) >> 1, pos = 1;
7         for (int i = best_l; i <= best_r; i++) {
8             if (dp[i][j - 1] + cost(i + 1, mid) < dp[mid][j]) {
9                 pos = i;
10                 dp[mid][j] = dp[i][j - 1] + cost(i + 1, mid);
11             }
12         }
13         dc(j, l, mid - 1, best_l, pos);
14         dc(j, mid + 1, r, pos, best_r);
15     }
16     void solve() {
17         for (int i = 1; i <= n; i++)
18             dp[i][1] = pre[i] * pre[i];
19         for (int j = 2; j <= k; j++)
20             dc(j, j, n, j - 1, n);
21     }
22 }

```

5.3 digit dp

```

1 // 原題：計算 [l, r] 中兩兩相鄰數字不同的數字數量
2 // 注意邊界
3 int dp[20][10]; // i 位數，尾數是 j
4 void init() {
5     for (int j = 0; j <= 9; j++) dp[1][j] = 1;
6     for (int i = 2; i <= 18; i++)
7         for (int now = 0; now <= 9; now++) // 這個位數的數字
8             for (int pre = 0; pre <= 9; pre++) // 上個位數的數字
9                 if (now != pre)
10                     dp[i][now] += dp[i - 1][pre];
11 }

```

```

12 int calc(int x) { // 小於等於 x 的數量
13     if (x == 0) return 0;
14     int arr[20] = {}, len = 0;
15     while (x > 0) arr[++len] = x % 10, x /= 10;
16     // 計算 len 位數的數量
17     int cnt = 0, pre = -1;
18     for (int idx = len; idx >= 1; idx--) {
19         for (int now = (idx == len); now < arr[idx]; now++)
20             if (now != pre)
21                 cnt += dp[idx][now];
22         if (arr[idx] == pre) break;
23         pre = arr[idx];
24         if (idx == 1) cnt++; // 特判
25     }
26     // 計算小於 len 位數的數量
27     for (int i = 1; i < len; i++)
28         for (int j = 1; j <= 9; j++)
29             cnt += dp[i][j];
30     return cnt;
31 }

```

5.4 dinic

```

1 struct Flow {
2     struct Edge {
3         // rc : residual capacity
4         int u, rc, rev;
5     };
6     vector<vector<Edge>> G;
7     vector<int> dis, it;
8     Flow(int n) : G(n), dis(n), it(n) {}
9     void add(int v, int u, int c) {
10         G[v].push_back({u, c, sz(G[u])});
11         G[u].push_back({v, 0, sz(G[v]) - 1});
12     }
13     int dfs(int v, int t, int f) {
14         if (v == t || f == 0) return f;
15         for (int &i = it[v]; i < sz(G[v]); i++) {
16             auto &[u, rc, rev] = G[v][i];
17             if (dis[u] != dis[v] + 1) continue;
18             int res = dfs(u, t, min(f, rc));
19             if (res <= 0) continue;
20             rc -= res;
21             G[u][rev].rc += res;
22             return res;
23         }
24         return 0;
25     }
26     int flow(int s, int t) {
27         int ans = 0;
28         for (int l = 30; l >= 0; l--) while (true) {
29             fill(all(dis), INT_MAX);
30             queue<int> Q;
31             Q.push(s);
32             dis[s] = 0;
33             while (!Q.empty()) {
34                 int v = Q.front(); Q.pop();
35                 for (auto [u, rc, rev] : G[v]) {
36                     if ((rc >> 1) <= 0 || dis[u] < INT_MAX) continue;
37                     dis[u] = dis[v] + 1;
38                     Q.push(u);
39                 }
40             }
41             if (dis[t] == INT_MAX) break;
42             fill(all(it), 0);
43             while (true) {
44                 int res = dfs(s, t, INT_MAX);
45                 if (res <= 0) break;
46                 ans += res;
47             }
48         }
49         return ans;
50     }
51 }
52 };

```

5.5 geometry

```

1 double eps = 0.000001;
2 pdd operator + (pdd a, pdd b) { return {a.X + b.X, a.Y + b.Y}; }
3 pdd operator - (pdd a, pdd b) { return {a.X - b.X, a.Y - b.Y}; }
4 double dot(pdd a, pdd b) { return a.X * b.X + a.Y * b.Y; }
5 double cross(pdd a, pdd b) { return a.X * b.Y - a.Y * b.X; }
6 int sign(double x) { return (fabs(x) < eps ? 0 : (x > 0 ? 1 : -1)); }
7 int ori(pdd a, pdd b, pdd c) { return sign(cross(b - a, c - a)); }
8 // c is between a and b
9 bool btw(pdd a, pdd b, pdd c) {

```

```

10 return (sign(cross(a - c, b - c)) == 0 && sign(dot(a - c, b -
11 c)) <= 0);
12 // ab is touch cd
13 bool banana(pdd a, pdd b, pdd c, pdd d) {
14     int a123 = ori(a, b, c), a124 = ori(a, b, d);
15     int a341 = ori(c, d, a), a342 = ori(c, d, b);
16     if (btw(a, b, c) || btw(a, b, d) || btw(c, d, a) || btw(c, d,
17         b))
18         return true;
19     return (a123 * a124 < 0 && a341 * a342 < 0);
20 }
21 vector<pdd> convex_hull(vector<pdd> pts) {
22     sort(pts.begin(), pts.end());
23     pts.resize(unique(pts.begin(), pts.end()) - pts.begin());
24     vector<pdd> hull(1, pts[0]);
25     for (int t = 0; t < 2; t++) {
26         int sz = hull.size();
27         for (int i = 1; i < (int)pts.size(); i++) {
28             while ((int)hull.size() > sz && ori(hull[(int)hull.size()
29                 - 2], hull.back(), pts[i]) < 0)
30                 hull.pop_back();
31             hull.push_back(pts[i]);
32         }
33         reverse(pts.begin(), pts.end());
34     }
35     hull.pop_back();
36     return hull;
37 }
38 double area(vector<pdd> pt) {
39     double cnt = 0.0;
40     for (int i = 0; i < (int)pt.size() - 1; i++)
41         cnt += cross(pt[i], pt[i + 1]);
42     cnt += cross(pt[(int)pt.size() - 1], pt[0]);
43     return fabs(cnt);
44 }
45 void solve() {
46     int n;
47     cin >> n;
48     vector<pdd> pts;
49     pts.resize(n);
50     for (int i = 0; i < n; i++)
51         cin >> pts[i].X >> pts[i].Y;
52     int ans = area(convex_hull(pts));
53     if (ans % 2 == 1)
54         cout << ans / 2 << ".5\n";
55     else
56         cout << ans / 2 << ".0\n";
57 }

```

5.6 line container

```

1 // only works for integer coordinates!! maintain max
2 struct Line {
3     mutable ll a, b, p;
4     bool operator<(const Line &rhs) const { return a < rhs.a; }
5     bool operator<(ll x) const { return p < x; }
6 };
7 struct DynamicHull : multiset<Line, less<>> {
8     static const ll kInf = 1e18;
9     ll Div(ll a, ll b) { return a / b - ((a ^ b) < 0 && a % b); }
10    bool isect(iterator x, iterator y) {
11        if (y == end()) { x->p = kInf; return 0; }
12        if (x->a == y->a) x->p = x->b > y->b ? kInf : -kInf;
13        else x->p = Div(y->b - x->b, x->a - y->a);
14        return x->p >= y->p;
15    }
16    void addline(ll a, ll b) { // ax + b
17        auto z = insert({a, b, 0}); y = z++; x = y;
18        while (isect(y, z)) z = erase(z);
19        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
20        while ((y = x) != begin() && (--x)->p >= y->p) isect(x,
21            erase(y));
22    }
23    ll query(ll x) {
24        auto l = *lower_bound(x);
25        return l.a * x + l.b;
26    }
27 };

```

5.7 mcmf

```

1 struct Flow {
2     struct Edge {
3         int u, rc, k, rv;
4     };
5     vector<vector<Edge>> G;
6     vector<int> par, par_eid;
7     Flow(int n) : G(n + 1), par(n + 1), par_eid(n + 1) {}
8     void add(int v, int u, int c, int k) {
9         G[v].push_back({u, c, k, (int)G[u].size()});

```

```

10         G[u].push_back({v, 0, -k, (int)G[v].size() - 1});
11     }
12     int spfa(int s, int t) {
13         fill(par.begin(), par.end(), -1);
14         vector<int> dis(par.size(), LONG_LONG_MAX);
15         vector<bool> in_q(par.size(), false);
16         queue<int> Q;
17         dis[s] = 0; in_q[s] = true;
18         Q.push(s);
19         while (!Q.empty()) {
20             int v = Q.front(); Q.pop();
21             in_q[v] = false;
22             for (int i = 0; i < (int)G[v].size(); i++) {
23                 auto [u, rc, k, rv] = G[v][i];
24                 if (rc > 0 && dis[v] + k < dis[u]) {
25                     dis[u] = dis[v] + k;
26                     par[u] = v;
27                     par_eid[u] = i;
28                     if (!in_q[u]) Q.push(u);
29                     in_q[u] = true;
30                 }
31             }
32         }
33         return dis[t];
34     }
35     pair<int, int> flow(int s, int t) {
36         int fl = 0, cost = 0, d;
37         while ((d = spfa(s, t)) < LONG_LONG_MAX) {
38             int cur = LONG_LONG_MAX;
39             for (int v = t; v != s; v = par[v])
40                 cur = min(cur, G[par[v]][par_eid[v]].rc);
41             fl += cur;
42             cost += d * cur;
43             for (int v = t; v != s; v = par[v]) {
44                 G[par[v]][par_eid[v]].rc -= cur;
45                 G[v][G[par[v]][par_eid[v]].rv].rc += cur;
46             }
47         }
48         return {fl, cost};
49     }
50 };

```

5.8 mo

```

1 struct MoSolver {
2     struct query {
3         int l, r, id;
4         bool operator< (const query &o) {
5             if (l / C == o.l / C) return (l / C) & 1 ? r > o.r : r <
6                 o.r;
7             return l / C < o.l / C;
8         }
9     };
10    int cur_ans;
11    vector<int> ans;
12    void add(int x) {
13        // do something
14    }
15    void sub(int x) {
16        // do something
17    }
18    vector<query> Q;
19    void add_query(int l, int r, int id) {
20        // [l, r]
21        Q.push_back({l, r, id});
22        ans.push_back(0);
23    }
24    void run() {
25        sort(Q.begin(), Q.end());
26        int pl = 0, pr = 0;
27        cur_ans = 0;
28        for (query &i : Q) {
29            while (pl > i.l)
30                add(a[--pl]);
31            while (pr < i.r)
32                add(a[pr++]);
33            while (pl < i.l)
34                sub(a[pl++]);
35            while (pr > i.r)
36                sub(a[pr--]);
37            ans[i.id] = cur;
38        }
39    }
40 };

```

5.9 number split

```

1 // dp[i][x] = 要將整數 x 拆成 i 堆的「組合數」
2 dp[i + 1][x + 1] += dp[i][x]; // 創造新的一堆
3 dp[i][x + i] += dp[i][x]; // 把每一堆都增加 1

```

5.10 sos dp

```
1 // 時間複雜度為  $O(n * 2^n)$ 
2 // 計算  $dp[i] = i$  所有 bit mask 子集的和
3 for (int i = 0; i < n; i++)
4     for (int mask = 0; mask < (1 << n); mask++)
5         if ((mask >> i) & 1)
6             dp[mask] += dp[mask ^ (1 << i)];
```

5.11 subset

```
1 for (int i = 0; i < (1 << n); i++){
2     for (int m = i; m; m = (m - 1) & i){
3         // m 是 i 的非空子集
4     }
5 }
```

5.12 xor basis

```
1 // 給一數字集合 S，求能 XOR 出多少數字
2 // 答案等於  $2^{\{basis\}}$  的大小
3 vector<int> basis;
4 void add_vector(int x) {
5     for (int v : basis) x = min(x, x ^ v);
6     if (x) basis.push_back(x);
7 }
8 // 給一數字集合 S，求能不能 XOR 出 x
9 bool check(int x) {
10     for (int v : basis) x = min(x, x ^ v);
11     return x;
12 }
13 // 給一數字集合 S，求 XOR 出最大的數字
14 int get_max() {
15     int ans = 0;
16     for (auto v : basis) ans = max(ans, ans ^ v);
17     return ans;
18 }
```

6 string

6.1 hash

```
1 const int p = 1e6 + 3;
2 const int mod = 1e9 + 7;
3 int hash(string &s) {
4     int res = 0;
5     for (int i = 0; i < s.size(); i++) {
6         res *= p;
7         res += s[i] - 'a' + 1;
8         res %= mod;
9     }
10    return res;
11 }
12 int pow_p[1000005];
13 vector<int> rh;
14 void build(string &s) {
15     rh.resize(s.size() + 1);
16     pow_p[0] = 1;
17     for (int i = 1; i <= s.size(); i++)
18         pow_p[i] = (pow_p[i - 1] * p) % mod;
19     rh[0] = s[0] - 'a' + 1;
20     for (int i = 1; i < s.size(); i++) {
21         rh[i] = rh[i - 1] * p + (s[i] - 'a' + 1);
22         rh[i] %= mod;
23     }
24 }
25 int query(int l, int r) {
26     int res = rh[r] - (l > 0 ? rh[l - 1] * pow_p[r - l + 1] : 0);
27     res = (res % mod + mod) % mod;
28     return res;
29 }
```

6.2 kmp

```
1 vector<int> F;
2 void build_failure_function(string &s) {
3     F.clear();
4     F.resize(s.size(), -1);
5     for (int i = 1; i < s.size(); i++) {
6         int j = F[i - 1];
7         while (j != -1 && s[i] != s[j + 1])
```

```
8             j = F[j];
9         if (s[i] == s[j + 1])
10             F[i] = j + 1;
11     }
12 }
13 void KMP_matching(string &a, string &b) {
14     for (int i = 0, j = -1; i < a.size(); i++) {
15         while (j != -1 && a[i] != b[j + 1])
16             j = F[j];
17         if (a[i] == b[j + 1])
18             j++;
19         if (j + 1 == b.size()) {
20             cout << "found a matching start at " << i - j << "\n";
21             j = F[j];
22         }
23     }
24 }
```

6.3 manacher

```
1 string s;
2 int extend(int l, int r, int N) {
3     int i = 0;
4     while (l - i >= 0 && r + i < N && s[l - i] == s[r + i])
5         i++;
6     return i;
7 }
8 int Longest_Palindromic_Substring(string &t) {
9     int N = t.length();
10    s.resize(2 * N + 1, '$');
11    for (int i = 0; i < N; i++)
12        s[2 * i + 1] = t[i];
13    N = 2 * N + 1;
14    vector<int> res;
15    res.resize(N, 0);
16    res[0] = 1;
17    for (int i = 1, mid = 0, R = 0; i < N; i++) {
18        int j = mid - (i - mid);
19        int lst = R + 1 - i;
20        if (i > R) {
21            res[i] = extend(i, i, N);
22            mid = i;
23            R = i + res[i] - 1;
24        } else if (res[j] == lst) {
25            res[i] = lst + extend(i - R, i + R, N);
26            mid = i;
27            R = i + res[i] - 1;
28        } else {
29            res[i] = min(res[j], lst);
30        }
31    }
32    int mx = -1, idx = -1;
33    for (int i = 0; i < N; i++) {
34        if (res[i] > mx) {
35            mx = res[i];
36            idx = i;
37        }
38    }
39    cout << "Longest Palindromic Substring Length is " << (mx + 1) / 2 << "\n";
40    for (int i = idx - res[idx] + 1; i <= idx + res[idx] + 1; i++)
41        if (i & 1)
42            cout << s[i];
43    cout << "\n";
44 }
```

6.4 string trie

```
1 struct node {
2     node* link[26] = {nullptr};
3     bool flag = false;
4     bool contain_key(char ch) { return (link[ch - 'a'] != nullptr); }
5     void put(char ch, node* _node) { link[ch - 'a'] = _node; }
6     node* get(char ch) { return link[ch - 'a']; }
7     void set_end() { flag = true; }
8     bool is_end() { return flag; }
9 };
10 struct Trie {
11     node* root;
12     Trie() { root = new node(); }
13     void insert(string s) { // insert a string
14         node* now = root;
15         for (int i = 0; i < (int)s.size(); i++) {
16             if (!now->contain_key(s[i])) now->put(s[i], new node());
17             now = now->get(s[i]);
18         }
19         now->set_end();
20     }
21     // return if there is a word of the given string
```

```

22 bool query(string s) {
23     node* now = root;
24     for (int i = 0; i < (int)s.size(); i++) {
25         if (!now->contain_key(s[i])) return false;
26         now = now->get(s[i]);
27     }
28     return now->is_end();
29 }
30 // return if there is any word that start with the given
    prefix
31 bool query_prefix(string s) {
32     node* now = root;
33     for (int i = 0; i < (int)s.size(); i++) {
34         if (!now->contain_key(s[i])) return false;
35         now = now->get(s[i]);
36     }
37     return true;
38 }
39 };

```

6.5 xor trie

```

1 struct node {
2     node* link[2];
3     bool contain_key(int val) { return (link[val] != nullptr); }
4     node* get(int val) { return link[val]; }
5     void put(int val, node* _node) { link[val] = _node; }
6 };
7 struct Trie {
8     node* root;
9     Trie() { root = new node(); }
10    void update(int x) {
11        node* now = root;
12        for (int i = 30; i >= 0; i--) {
13            if (x & (1 << i)) {
14                if (!now->contain_key(1)) now->put(1, new node());
15                now = now->get(1);
16            } else {
17                if (!now->contain_key(0)) now->put(0, new node());
18                now = now->get(0);
19            }
20        }
21    }
22    int query(int x) { // query max xor with number x
23        node* now = root;
24        int mx = 0;
25        for (int i = 30; i >= 0; i--) {
26            int nowbit = (x >> i) & 1;
27            int target = nowbit ^ 1;
28            if (now->contain_key(target)) {
29                mx += (1 << i);
30                now = now->get(target);
31            } else {
32                now = now->get(nowbit);
33            }
34        }
35        return mx;
36    }
37 };

```

6.6 z

```

1 vector<int> Z;
2 void calculate_Z(string &s) {
3     Z.clear();
4     Z.resize(s.size(), 0);
5     int l = 0, r = 0;
6     for (int i = 1; i < s.size(); i++) {
7         if (i <= r)
8             Z[i] = min(Z[i - 1], r - i + 1);
9         while (i + Z[i] < s.size() && s[i + Z[i]] == s[Z[i]])
10            Z[i]++;
11         if (i + Z[i] - 1 > r) {
12             l = i; r = i + Z[i] - 1;
13         }
14     }
15 }
16 void Z_matching(string &a, string &b) {
17     string res = "";
18     int i = 0;
19     for (char ch : b)
20         res[i++] = ch;
21     res[i++] = '_';
22     for (char ch : a)
23         res[i++] = ch;
24     calculate_Z(res);
25     for (int i = 0; i < Z.size(); i++) {
26         if (Z[i] == b.size())
27             cout << "found a matching start at " << i - b.size() - 1
28                 << "\n";
29     }
30 }

```

```

29 }

```

7 tree

7.1 hld

```

1 const int N = 100005;
2 vector<int> G[N];
3 struct HLD {
4     vector<int> pa, sz, depth, mxson, topf, id;
5     int n, idcnt = 0;
6     HLD(int _n) : n(_n), pa(_n + 1), sz(_n + 1), depth(_n + 1),
            mxson(_n + 1), topf(_n + 1), id(_n + 1) {}
7     void dfs1(int v = 1, int p = -1) {
8         pa[v] = p; sz[v] = 1; mxson[v] = 0;
9         depth[v] = (p == -1 ? 0 : depth[p] + 1);
10        for (int u : G[v]) {
11            if (u == p) continue;
12            dfs1(u, v);
13            sz[v] += sz[u];
14            if (sz[u] > sz[mxson[v]]) mxson[v] = u;
15        }
16    }
17    void dfs2(int v = 1, int top = 1) {
18        id[v] = ++idcnt;
19        topf[v] = top;
20        if (mxson[v]) dfs2(mxson[v], top);
21        for (int u : G[v]) {
22            if (u == mxson[v] || u == pa[v]) continue;
23            dfs2(u, u);
24        }
25    }
26    int path_query(int a, int b) {
27        int res = 0;
28        while (topf[a] != topf[b]) {
29            if (depth[topf[a]] < depth[topf[b]]) swap(a, b);
30            res = max(res, query(id[topf[a]], id[a])); // query : L =
                id[topf[a]], r = id[a]
31            a = pa[topf[a]];
32        }
33        if (depth[a] < depth[b]) swap(a, b);
34        res = max(res, query(id[b], id[a])); // query : L = id[b],
                r = id[a]
35        return res;
36    }
37 };

```